ICT2142 E- Business Systems

Lecture 03

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E-Commerce Infrastructure and Technologies

Objectives

- Explain the Internet technologies that provide the foundation for e-commerce.
- Describe the role of DNS, TCP/IP, HTTP/HTTPS, browsers, and web technologies in online transactions.
- Explain web servers, hosting models, scalability, availability, and basic deployment considerations.
- Describe Web APIs and how third-party services support payments, logistics, authentication, and other business functions.
- Explain mobile commerce (m-commerce) architecture, technologies, benefits, and challenges.

Internet Technologies Enabling E-Commerce

- The Internet provides the communication infrastructure through which customers, businesses, banks, payment providers, and logistics providers interact.
- Core technologies include:
  - ✓ Internet Protocol (IP)
  - ✓ DNS
  - ✓ TCP/IP
  - ✓ HTTP/HTTPS, web browsers, web servers, databases, and cloud services.

- An e-commerce transaction typically involves several layers.

Example: A customer opens an e-commerce site of an online shop, searches for a product, adds it to a cart, and submits an order.

DNS finds the destination server; HTTPS protects communication; the application processes business logic; data is retrieved or stored; external APIs may complete payment or delivery operations.



- E-commerce systems must provide security, availability, acceptable response time, scalability, and reliable data exchange.

Domain Name System (DNS)

- DNS translates human-readable domain names (e.g., shop.example.com) into IP addresses used by network devices.
- DNS is important to e-commerce because customers normally access stores using domain names rather than numeric IP addresses.

- DNS also supports availability and traffic management through techniques such as multiple records, routing policies, and failover configurations.
- Security consideration: DNS should be managed carefully to reduce risks such as domain hijacking and DNS spoofing.

TCP/IP: The Foundation of Internet Communication

TCP/IP is the fundamental communication system that allows data to travel from one device to another across the Internet.

TCP – Transmission Control Protocol

- TCP is responsible for reliable delivery of data.
- When a large amount of information needs to be transmitted, TCP divides it into smaller pieces called **segments** and helps ensure that they arrive correctly and in the proper order.

IP– Internet Protocol

- IP is responsible for addressing and routing data.
- Every device communicating over an IP network has an **IP address**. IP uses these addresses to determine where data should be sent.

HTTP vs HTTPS

HTTP and **HTTPS** are protocols used for communication between a **web browser (client)** and a **web server**.

Feature	HTTP	HTTPS
Full form	Hypertext Transfer Protocol	Hypertext Transfer Protocol Secure
Security	Not encrypted	Encrypted
Encryption	No TLS encryption	Uses **TLS (Transport Layer Security)**
Data protection	Data can potentially be intercepted or modified	Data is protected during transmission
Authentication	Does not provide server authentication through TLS	TLS provides server authentication
Data integrity	No TLS-based integrity protection	Provides integrity protection
URL example	http://example.com	https://example.com
E-commerce use	Not suitable for sensitive transactions	**Essential for e-commerce**

Client–Side vs Server–Side Processing

Client–Side

- Runs mainly in the customer's browser/device.
- Examples: UI rendering, form validation, interactive product filters.
- Advantages: fast interaction and reduced server workload.
- Limitation: cannot be trusted for security–sensitive business decisions.

Server–Side

- Runs on web/application servers.
- Examples: authentication, order processing, pricing rules, inventory updates.
- Advantages: centralized control and access to protected resources.
- Security principle: validate important inputs and business rules on the server.

Web Browsers and Web Technologies

- **HTML:** Defines the structure of a web page.

  Forms, links, product information, images, tables, and semantic elements.

- **CSS:** Controls presentation and layout.

  Responsive design helps pages adapt to desktops, tablets, and smartphones.

- **JavaScript:** Adds interaction and dynamic behavior.

  Supports client-side validation, asynchronous requests, cart updates, and rich user interfaces.

Web Application Architecture



- Presentation layer: displays products, carts, forms, accounts, and order information.
- Application layer: implements authentication, shopping cart logic, pricing, promotions, inventory, and order management.
- Data layer: stores customer, product, inventory, order, and transaction-related data.
- External services: payment gateways, shipping/logistics, email/SMS, identity providers, analytics, and other APIs.

Caching and Content Delivery Networks (CDNs)

- Caching stores frequently requested data temporarily so it can be served faster.
- Browser cache: stores selected resources on the user's device.
- Server/application cache: stores frequently used data or computed results.
- CDN: distributes static content such as images, CSS, JavaScript, and videos through geographically distributed edge locations.
- Benefits: lower latency, reduced origin-server load, improved scalability, and better user experience.
- E-commerce example: product images and static assets can be delivered from a nearby CDN edge while orders continue to be processed by the application server.

Web Servers and Hosting Options

- A web server receives HTTP/HTTPS requests and returns web resources or forwards requests to application components.

  Examples of web-server software include Apache HTTP Server,

  Nginx, and Microsoft IIS.

- Static websites can be served directly from a web server or object storage/CDN.
- Dynamic e-commerce applications normally involve an application runtime and a database behind the web server.
- Hosting choice affects cost, performance, scalability, security responsibilities, availability, and maintenance effort.

Static vs Dynamic Web Content

Static Web Pages: Same stored resource can be delivered to many users.

- ✓ Examples: HTML pages, images, CSS, JavaScript files.
- ✓ Easy to cache and distribute through CDNs.
- ✓ Suitable for informational pages and static product assets.

Dynamic Web Pages: Generated or customized based on data, user, session, or request.

- ✓ Examples: shopping cart, account page, stock availability, personalized recommendations.
- ✓ Usually requires application processing and database access.
- ✓ Requires careful performance and security management.

Web Hosting Options

- Shared Hosting
- VPS (Virtual Private Servers)
- Dedicated Servers
- Cloud Based Servers

Cloud Hosting for E-Commerce

- Cloud platforms provide compute, storage, networking, databases, monitoring, security, and other managed services.
- Infrastructure can be scaled horizontally by adding more application instances.
- Load balancers distribute requests across multiple application servers.
- Auto-scaling can increase or decrease capacity based on demand.
- Managed databases reduce some administrative tasks such as backups, patching, and high-availability configuration.
- Key trade-offs: cost control, vendor dependency, configuration complexity, security, data location, and operational skills.

Scalability, Availability and Performance

Scalability

- Ability to handle increasing workload.
- Vertical scaling: increase resources of a server.
- Horizontal scaling: add more servers/instances.

Availability

- Ability to remain accessible and operational.
- Redundancy, health checks, failover, backups, and monitoring improve resilience.

Scalability, Availability and Performance

Performance

- How quickly and efficiently the system responds.
- Caching, CDN, database optimization, efficient APIs, and appropriate infrastructure can reduce latency.

Web APIs and Third-Party Integrations

- An API (Application Programming Interface) defines how software components communicate and exchange data.
- E-commerce businesses rarely build every supporting service themselves; they integrate specialized external providers.
- Common integrations: payment gateways, delivery/logistics, identity and authentication, email/SMS, maps, analytics, tax, and fraud detection.
- APIs reduce development effort and enable businesses to connect different systems.
- Integration quality affects reliability, security, customer experience, and operational continuity.

Payment Gateway Integration

- A payment gateway connects an e-commerce application with payment processing infrastructure.
- Integration approaches may include hosted payment pages, redirect-based flows, embedded components, or direct API integrations.
- **Important concepts:** transaction ID, authorization, capture, refund, cancellation, webhook/callback, reconciliation.
- **Security considerations:** HTTPS, authentication credentials, access control, secure secret storage, validation, fraud controls, logging, and compliance requirements.
- Good design should handle timeouts, duplicate requests, failed payments, delayed notifications, and retry scenarios.

Logistics and Delivery APIs

- Logistics APIs allow an e-commerce system to exchange information with delivery/courier providers.
- Common functions: shipping-rate calculation, address validation, shipment creation, tracking, delivery status, and proof of delivery.

Example workflow: order confirmed → shipment created → tracking number returned → tracking updates received → customer notified.

- Webhooks can allow the logistics provider to notify the e-commerce system when shipment status changes.
- Benefits: automated order fulfillment, real-time tracking, fewer manual operations, and better customer visibility.

Authentication and Identity APIs

- Authentication answers: “Who is the user?” Authorization answers: “What is the user allowed to do?”
- E-commerce systems may use their own identity service or integrate with an external identity provider.
- Common technologies/concepts: sessions, cookies, access tokens, OAuth 2.0, OpenID Connect, and multi-factor authentication.
- Social login can reduce friction by allowing customers to authenticate using an existing identity provider.
- **Security practices:** strong credential protection, short-lived tokens where appropriate, secure cookies, least privilege, and protection against session/token theft.

Mobile Commerce (M-Commerce)

- M-commerce is the buying, selling, payment, and other commercial activities conducted using mobile devices.
- Examples: mobile shopping websites, mobile apps, mobile banking, digital wallets, QR-based payments, and in-app purchases.
- Mobile users expect fast loading, simple navigation, touch-friendly controls, secure authentication, and reliable payment experiences.
- M-commerce depends on mobile networks, smartphones, browsers/apps, APIs, cloud infrastructure, payment services, and location/notification capabilities.

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Mobile Commerce Technologies

Responsive Web

One website adapts to different screen sizes.

HTML + CSS + JavaScript.

Accessible through a mobile browser.

Native Apps

Built specifically for a mobile platform.

Can access device capabilities through platform APIs.

Useful for rich, frequent interactions.

Cross-Platform

Shared codebase can target multiple platforms.

Can reduce development effort.

Uses frameworks and platform bridges.

Mobile Payments

Digital wallets, QR payments, cards, bank-based payments, and provider APIs.

Security and user authentication are critical.

Activity

1. What is RESTful Web APIs?

2. What are the Mobile UX and performance considerations?

3. Describe how to maintain security across E-Commerce infrastructure on network, application, data and integration?

Thank you… !